

# Final Project Report

## 1. Introduction and Objective of the Analysis

Online movie platforms such as IMDb play an important role in how audiences discover and evaluate films. IMDb ratings are widely used as a measure of movie quality and often influence viewers' decisions about which movies to watch. As a result, understanding what factors are associated with high IMDb ratings can provide valuable insights into how movies are perceived by audiences.

The aim of this analysis is to explore whether selected movie characteristics are associated with a higher probability of achieving a very high IMDb rating. Instead of predicting the exact rating value, the focus is aimed at modeling the likelihood of success, defined as an IMDb rating of at least 8. This threshold is commonly considered an indicator of particularly well-received movies.

Using data from the IMDb Top 1000 movies dataset, the analysis investigates the relationship between movie success and selected characteristics such as popularity, runtime, and release year. The main goal is to identify which of these factors are most strongly associated with the probability of IMDb success and to provide an interpretable, data-driven explanation of these relationships.

## 2. Dataset Description and Preparation, Context

The analysis is based on the IMDb Top 1000 Movies dataset, which contains information about some of the highest-rated and most popular movies listed on IMDb. The dataset includes various features describing each movie, such as audience ratings, number of votes, runtime, release year, and additional descriptive information.

After basic data cleaning and preparation, the final dataset consists of 999 movies. One observation was removed due to missing release year after converting `Released_Year` to

numeric. All variables used in the analysis were checked for reliability and consistency before proceeding with the exploratory analysis and model estimation. After cleaning, the dataset provides a reasonable basis for analyzing factors associated with movie success, as it includes a diverse set of films released across different decades and genres, while maintaining a consistent level of data quality.

The IMDb Top 1000 Movies dataset is particularly well suited for this analysis because it focuses on movies that have received significant audience attention. This allows for a meaningful comparison between highly successful movies and those that, while still popular, did not achieve the highest ratings.

### **3. Short Research Context**

Movie ratings are commonly used as an equivalent for audience satisfaction and quality. It is commonly suggested that factors such as popularity and audience engagement are closely related to how movies are evaluated by viewers. Movies that attract more attention are more likely to be widely discussed, rated, and remembered.

At the same time, characteristics such as runtime and release period may also influence audience perception, as longer movies often allow for more complex storytelling, while older movies may benefit from a “classic” status. These observations motivate a closer look at how selected movie characteristics may be related to very high IMDb ratings.

### **4. Variables Definition**

- Dependent Variable

The dependent variable in this analysis is movie success. Success is defined as a binary variable that takes the value of 1 if a movie has an IMDb rating of at least 8, and 0 otherwise. I chose this threshold because ratings of 8 or higher are commonly considered as an indicator of exceptional audience reception on IMDb.

Using a binary definition of success allows the analysis to focus on the probability of a movie achieving a very high rating, rather than modeling small differences in rating values. This

approach leads to clearer interpretation and is well suited for the applied modeling framework used in this project.

- **Independent Variables**

The analysis includes three independent variables that are expected to be related to movie success: popularity, runtime, and release year.

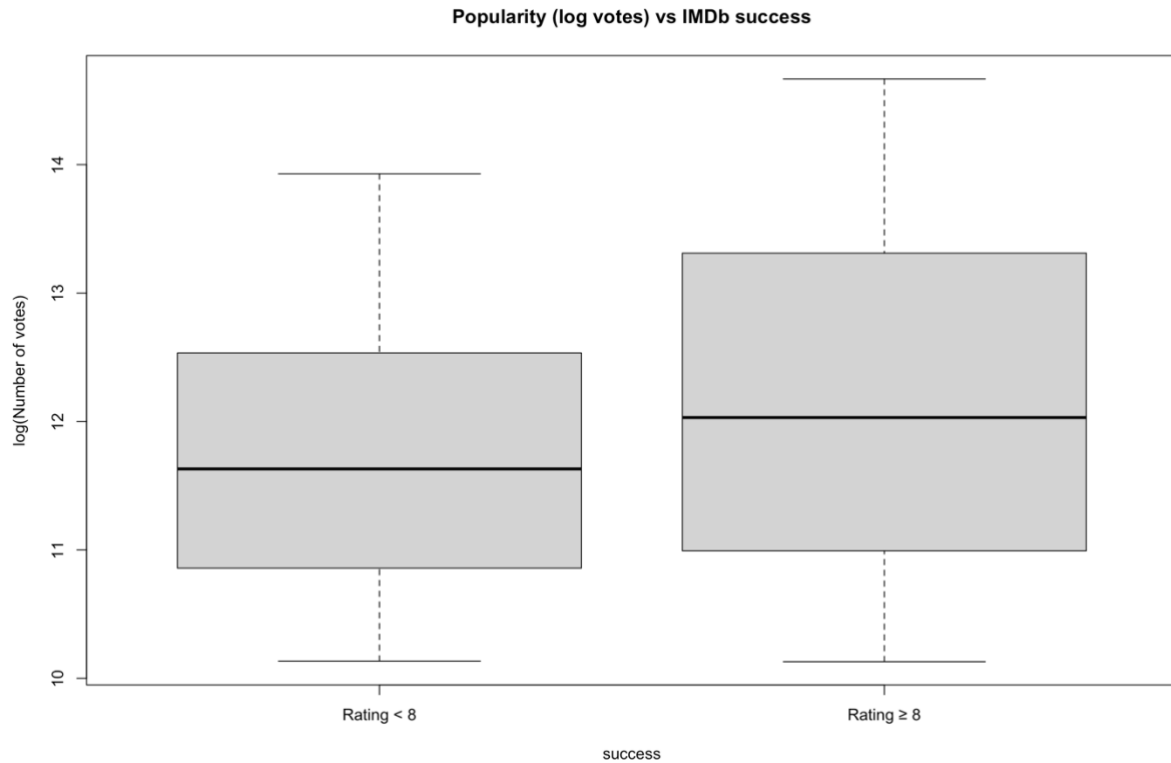
Popularity is measured by the number of votes a movie has received on IMDb. Since the distribution of the number of votes is highly skewed, I applied a logarithmic transformation. This transformation reduces the influence of extremely popular movies and allows for a more stable and interpretable relationship between popularity and success.

Runtime represents the duration of a movie in minutes. Longer movies may allow for more detailed storytelling and character development, which could influence audience perception and ratings.

Release year indicates the year in which a movie was released. This variable captures potential differences between older and more recent movies, such as changes in audience preferences or the presence of highly valued classic films.

## **5. Exploratory Data Analysis (EDA)**

I conducted Exploratory Data Analysis to gain an initial understanding of the relationships between movie success and selected explanatory variables. The goal of this step was to identify visible patterns and differences between successful and non-successful movies before applying a formal statistical model.



*Figure 1. Distribution of movie popularity (log-transformed number of votes) for successful and non-successful movies*

As shown in Figure 1, successful movies tend to receive considerably more votes. This pattern is visible in the distribution of the log-transformed number of votes and suggests that popularity may play an important role in explaining very high IMDb ratings.

Smaller differences can also be observed for other characteristics. Successful movies are, on average, slightly longer than non-successful movies, although the overlap between the two groups remains considerable. In addition, successful movies tend to be released earlier, which may reflect the presence of highly valued classic films among the top-rated movies on IMDb.

Overall, the exploratory analysis indicates that popularity is likely to be the strongest factor associated with movie success, while runtime and release year may have a weaker but still meaningful relationship. Based on these observations, all three variables were included in the further regression analysis.

## **6. Model Choice and Assumptions**

To analyze the relationship between movie characteristics and the probability of success, I used a logistic regression model. This choice was motivated by the binary nature of the dependent variable, which indicates whether a movie is classified as successful or not. Logistic regression is well suited for this type of outcome, as it directly models probabilities and ensures that predicted values lie between 0 and 1.

A linear regression model was not appropriate in this context, since it is designed for continuous outcomes and may produce predictions outside the valid probability range. In contrast, logistic regression allows for a clear and interpretable relationship between explanatory variables and the likelihood of success.

I considered several basic model assumptions before interpreting the results. In particular, the potential presence of multicollinearity among the independent variables was examined. I assessed multicollinearity using VIF; all VIF values were close to 1, indicating no meaningful multicollinearity. It suggests that the estimated effects can be interpreted independently. Overall, the chosen model is consistent with the structure of the data and aligns well with the patterns observed during EDA.

## **7. Model Estimation and Interpretation**

The logistic regression model was estimated using movie popularity, runtime, and release year as explanatory variables. The results confirm the patterns observed during exploratory data analysis and provide a more formal assessment of the relationships between these variables and the probability of movie success.

| Variable               | Odds Ratio | 95% CI       |
|------------------------|------------|--------------|
| Popularity (log votes) | 1.60       | [1.41, 1.83] |
| Runtime                | 1.02       | [1.01, 1.02] |
| Release year           | 0.98       | [0.97, 0.98] |

*Table 1. Estimated odds ratios and 95% confidence intervals from the logistic regression model.*

Table 1 presents the estimated odds ratios and confidence intervals for the explanatory variables included in the model. Popularity, measured as the log-transformed number of IMDb votes, clearly appeared as the strongest predictor of movie success in the model. An increase in popularity is associated with a vast growth in the chances of a movie achieving an IMDb rating of at least 8. This result indicates that movies that attract more audience attention are considerably more likely to be classified as successful.

Runtime also shows a positive relationship with success, although its effect is noticeably smaller than that of popularity. Longer movies tend to have slightly higher possibilities of success, suggesting that runtime may play a secondary role in shaping audience evaluations. In contrast, release year has a negative association with success, meaning that more recent movies are, on average, less likely to achieve very high IMDb ratings compared to older films. This effect may reflect the presence of well-established classic movies among the highest-rated titles.

Overall, the model results highlight that while multiple factors contribute to IMDb success, popularity is by far the most influential characteristic. Runtime and release year provide additional explanatory power but do not play the central role of audience engagement reflected in the number of votes.

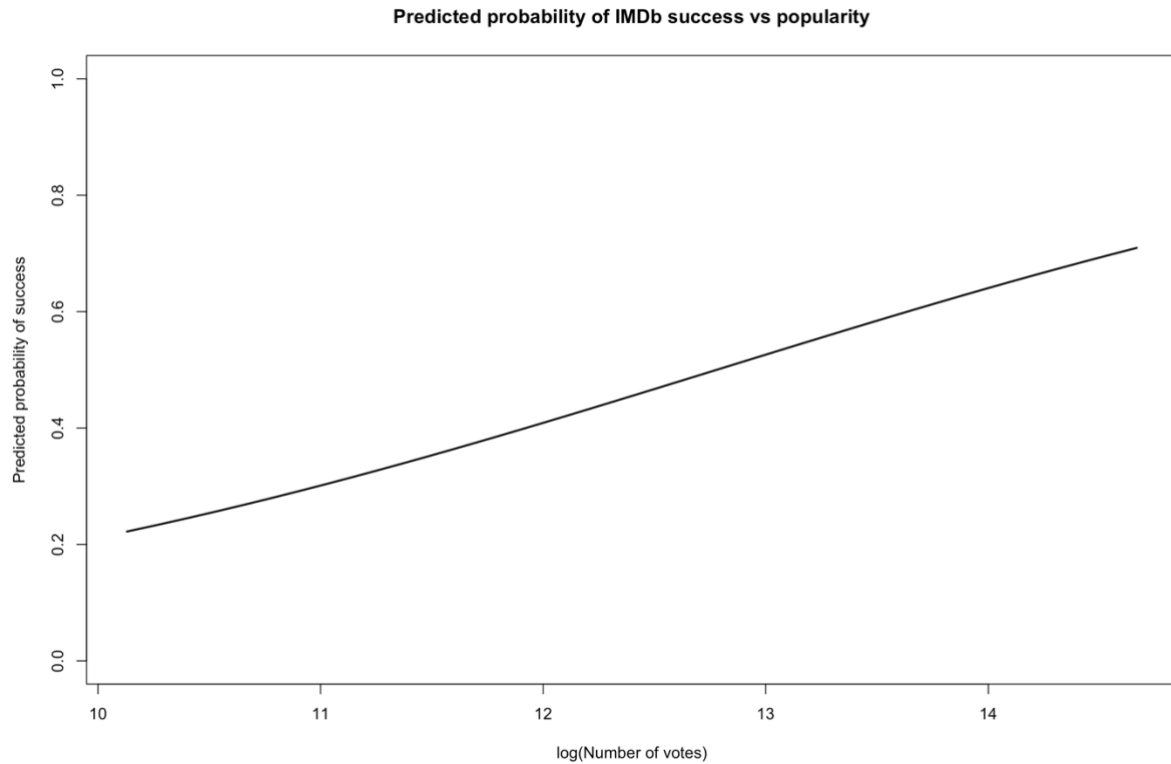
## 8. Model Performance

To evaluate the performance of the logistic regression model, I studied basic classification metrics. These measures provide an overview of how well the model distinguishes between successful and non-successful movies. Movies were classified as successful when the predicted probability was greater than or equal to 0.5.

The overall accuracy of the model is approximately 66%, meaning that about two thirds of the movies are correctly classified. The sensitivity of the model, which measures the ability to correctly identify successful movies, is around 57%. This indicates that the model is moderately effective in detecting movies that achieve very high IMDb ratings. The specificity, which reflects the ability to correctly identify non-successful movies, is approximately 75%, suggesting that the model performs better in recognizing movies without success.

Overall, these results indicate that the model has reasonable predictive power, considering the simplicity of the selected variables. While it is not intended as a highly optimized predictive tool, the model provides useful insight into the factors associated with IMDb success.

## 9. Results Visualization



*Figure 2. Predicted probability of IMDb success as a function of movie popularity*

To complement the numerical results of the logistic regression model, I created an appealing visualization. Figure 2 illustrates how the predicted probability of movie success increases with movie popularity.

The visualization reveals a clear and increasing relationship between these two variables. For movies with relatively low popularity, corresponding to approximately 20 thousand votes, the predicted probability of success is around 22%. As popularity increases to about 250 thousand votes, this probability rises to roughly 45%. For very popular movies with more than two million votes, the predicted probability of success reaches approximately 70%.

This visualization provides an intuitive interpretation of the model results by translating statistical estimates into probabilities that are easy to understand. It strengthens the conclusion that popularity plays a central role in determining whether a movie achieves very high IMDb ratings.



## **10. Conclusions**

The aim of this analysis was to identify which movie characteristics are associated with a higher probability of achieving a very high IMDb rating. Using data from the IMDb Top 1000 Movies dataset and a logistic regression framework, the study focused on popularity, runtime, and release year as key explanatory variables.

The results clearly indicate that movie popularity is the most important factor associated with IMDb success. Movies that receive more audience attention, reflected by a higher number of votes, have significantly higher odds and predicted probabilities of achieving an IMDb rating of at least 8. While runtime and release year also show statistically meaningful relationships with success, their effects are noticeably weaker and should be interpreted as secondary.

Overall, the results suggest that audience engagement plays a central role in explaining why some movies achieve very high IMDb ratings. Although additional movie characteristics may also contribute to success, popularity stands out as the main factor in explaining why some movies achieve outstanding audience reception.

## **11. Limitations**

This analysis has several limitations that should be considered when interpreting the results. First, the dataset is restricted to the IMDb Top 1000 movies, which already represent relatively popular and well-received films. As a result, the findings may not fully generalize to less popular or lower-rated movies.

Second, the analysis focuses on a limited set of explanatory variables. While popularity, runtime, and release year capture important aspects of movie success, other potentially relevant factors such as genre, budget, marketing intensity, or director characteristics were not included. Involving such variables could provide additional insights into the determinants of IMDb ratings.

Finally, it should be emphasized that the results describe associations rather than causal relationships. Although strong relationships between popularity and success are observed, the analysis does not establish whether popularity causes higher ratings or is simply correlated with them.

## **12. Use of AI tools (transparency statement)**

I used AI to assist me in minor coding decisions (log transformation for the number of votes, creation of the prediction grid for the probability plot, and calculation of basic classification metrics - confusion matrix, accuracy, sensitivity, specificity) and while reviewing this report. All results were checked and interpreted by me.

### **Dataset source:**

- <https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows?resource=download>